



A Multi-Feature Computational Framework for Protein–Protein Interaction Site Prediction

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1. Introduction

Biology Concepts · GNN and Geometric Symmetry

2. Literature Review

8 Papers Reviewed · Part 1 and Part 2

3. Research Gaps

PLM Dominance · Biophysical Supervision · Unbound Collapse

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7. Conclusion

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Protein-Protein Interaction (PPI)

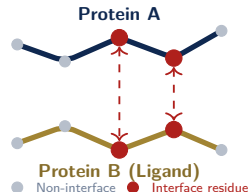
Two proteins bind via non-covalent forces governing **signal transduction**, **enzymatic inhibition**, and **immune response**.

Residue

Single amino acid unit in a protein chain. Each residue $i \in \{1, \dots, L\}$ is labelled *interface* ($y_i = 1$) or *non-interface* ($y_i = 0$). Only **~10–15%** of residues are interface.

Ligand

The binding *partner* protein (or small molecule) whose presence defines the interaction site on the target.



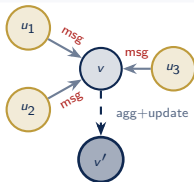
Task: Binary node classification – predict interface vs. non-interface per residue, under severe class imbalance (~10–15% positive labels).

Message Passing (GNN)

Each node v aggregates neighbour features and updates:

$$\mathbf{h}_v^{(\ell+1)} = \phi\left(\mathbf{h}_v^{(\ell)}, \bigoplus_{u \in \mathcal{N}(v)} \psi(\mathbf{h}_v^{(\ell)}, \mathbf{h}_u^{(\ell)})\right)$$

Each protein is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes $\mathcal{V}$ are residues and edges $\mathcal{E}$ connect spatially neighbouring residues within a distance threshold.



Rotation Equivariance vs. Invariance

Equivariance

$$f(\mathbf{R}\mathbf{x}) = \mathbf{R}f(\mathbf{x})$$

Output *rotates* with input

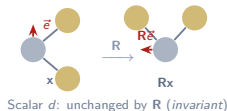
Coord updates in EGNN

Invariance

$$f(\mathbf{R}\mathbf{x}) = f(\mathbf{x})$$

Output *unchanged*

Scalar features (distances)



EGNN: equivariant coord updates
+ invariant scalar features.

Paper	Contribution	Key Limitation
EquiPPIS ^[1] Roche et al. 2023 <i>E(3) Equivariant GNN for Robust and Accurate PPIS</i>	E(3)-equivariant GNN; coordinate vector updates alongside PSSM/HMM features; robust on AlphaFold2 predicted structures	Classical MSA evolutionary profiles only; no protein language model (PLM) embeddings
PeSTo ^[2] Krapp et al. 2023 <i>Parameter-Free Geometric Deep Learning for Protein Binding Interfaces</i>	Parameter-free geometric transformer; raw atomic coordinates + element types; fast multi-partner interface prediction	Completely omits evolutionary profiles (PSSM, HMM, PLM)
E(Q)AGNN ^[3] Animesh et al. 2025 <i>Attention Enhanced Equivariant GNN for PPIS Prediction</i>	Attention-aggregated message passing + Geometric Vector Perceptrons (GVPs); directional vector equivariance	Evaluated on static bound crystals only; high compute overhead; no PLM integration
EGCPPIS ^[4] Sun et al. 2025 <i>Hierarchical Equivariant Graph Representations with Contrastive Integration for PPIS</i>	Hierarchical atom-level + residue-level graphs; contrastive cross-granularity learning; gated multi-head attention	High preprocessing latency from dual-tier graph construction; no PLM support

Paper	Contribution	Key Limitation
IIGRL [5] Zhao et al. 2024 <i>Intra-Inter Graph Representation Learning for PPI Binding Sites Prediction</i>	Intra-protein mutual-info graph + inter-protein affinity propagation network for paired binding	Requires paired complex at test time; partner-independent screening not possible
GTE-PPIS [6] Wang et al. 2025 <i>GTE-PPIS: Graph Transformer and EGNN for Protein–Protein Interaction Site Prediction</i>	Dual-branch EGNN (10L) + Graph Transformer (4L); 61D classical features; ensembled logits; SOTA on bound benchmarks	Degrades on unbound targets (AUPRC ~0.311–0.343); no cross-branch consensus; no PLM
DHEG [7] Li et al. 2026 <i>Predicting PPI Sites via Dynamic Perception in Hierarchical E(n)-Equivariant Graph</i>	Dynamic feature-perception in hierarchical E(n)-equivariant graph; first PLM audit showing naive ESM-2 concatenation fails (MCC 0.086)	No explicit biophysical gating (RSA, B-factor) to govern modality weighting
ESM-2 [8] Lin et al. 2023 <i>Evolutionary-Scale Prediction of Atomic-Level Protein Structure with a Language Model</i>	Transformer PLM (up to 15B params); encodes structural constraints and co-evolutionary patterns from sequence alone	Standalone embeddings poor at PPIS discrimination; naive fusion degrades geometric baselines

Gap 1 – PLM Dominance Dilemma

1280D ESM-2 embeddings overpower 61D structural features $\Rightarrow$ sequence overfitting, geometric discrimination lost.

MCC 0.086 with standalone ESM-2 [7]; AUPRC drop on bound benchmarks [6].

Gap 3 – Conformational Generalisation Collapse

SOTA models degrade severely on unbound structures. GTE-PPIS [6]: AUPRC ~ 0.311 on UBtest_31-6.

Induced-fit binding shifts backbone coordinates; coordinate-only GNNs fail without PLM grounding.

Gap 2 – No Joint Biophysical Supervision

No published PPIS framework uses joint RSA + B-factor supervision to determine per-residue geometry vs. PLM weighting.

Gap 4 – Unconstrained Latent Drift

Dual-branch models (EGNN + GT) optimise independently until late logit averaging, diverging into conflicting latent spaces under multi-modal inputs.

Formal Graph Formulation

Per-residue PPIS as **node-level binary classification** on graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with L residues. Each node i combines three complementary modalities:

1. **3D geometry:** PSSM (20D), HMM (20D), DSSP (14D), coordinates $\mathbf{x}_i \in \mathbb{R}^3$.
2. **PLM semantics:** ESM-2 [8] 1280D embeddings $\mathbf{p}_i$ (invariant to coordinate shifts).
3. **Biophysical state:** $\Phi_i = (\text{RSA}_i, B_{\text{norm},i})$ (solvent accessibility and flexibility).

Objective: Learn $f_\theta(\mathcal{G}) \rightarrow \hat{\mathbf{y}} \in [0, 1]^L$ under severe class imbalance ($\sim 10\text{--}15\%$ positive).

Challenge 1 – PLM Dominance

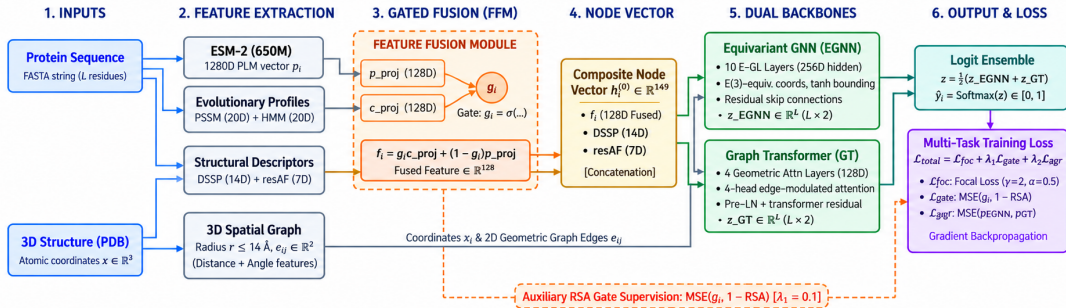
1280D PLM features dwarf 61D classical descriptors, causing representations to collapse toward sequence-only features.

Challenge 2 – Conformational Shift

Backbone shifts during binding cause pure coordinate-based GNNs to degrade on unbound structures.

Challenge 3 – Latent Drift

Uncoupled EGNN and GT branches diverge in latent feature space without cross-branch regularization.



Pipeline Flow: Raw Sequences & Coordinates $\rightarrow$ Multi-Modal Feature Extraction $\rightarrow$ Adaptive FFM Gating $\rightarrow$ Composite Node $h_i^{(0)}$ $\rightarrow$ Parallel Dual Backbones $\rightarrow$ Logit Ensembling & Multi-Task Loss



1. Gated Feature Fusion (Challenge 1 and Challenge 2)

- A learned gate $g_i \in [0, 1]$ decides, per residue, how much to trust classical features vs. ESM-2: $\mathbf{f}_i = g_i \mathbf{c}'_i + (1 - g_i) \mathbf{p}'_i$
- Gate is guided by exposure (RSA): buried $\Rightarrow$ classical, exposed $\Rightarrow$ ESM-2.
- DSSP descriptors skip the gate.

2. Dual-Branch Backbone (from GTE-PPIS) ^[6]

- **EGNN** (10 layers): local 3D geometry.
- **Graph Transformer** (4 layers): long-range context.
- Outputs are averaged into one prediction.

3. Training Loss (Challenge 3)

- **Focal loss**: focuses on rare interface residues.
- **Gate loss**: guides the gate using RSA.
- **Agreement loss**: keeps the two branches' predictions consistent.

Feature Fusion Module (Resolves Challenge 1 and Challenge 2)

Project and normalise both modalities, then gate:

$$\mathbf{c}'_i = \text{LN}(W_c \mathbf{c}_i), \quad \mathbf{p}'_i = \text{LN}(W_p \mathbf{p}_i)$$

$$g_i = \sigma(\mathbf{w}_g^\top [\mathbf{c}'_i \parallel \mathbf{p}'_i] + b_g)$$

$$\mathbf{f}_i = g_i \mathbf{c}'_i + (1 - g_i) \mathbf{p}'_i$$

LayerNorm prevents high-dimensional PLM scale overpowering geometry.

Biophysical Gate Supervision:

$$\tau_i = 1 - [\alpha_{\text{rsa}} \cdot \text{RSA}_i + (1 - \alpha_{\text{rsa}}) \cdot \beta_i]$$

$$\mathcal{L}_{\text{bio}} = \frac{1}{N} \sum_i (g_i - \tau_i)^2$$

Buried (RSA $\rightarrow 0$): $\tau \rightarrow 1 \Rightarrow$ prioritize geometry.

Surface (RSA $\rightarrow 1$): $\tau \rightarrow 0 \Rightarrow$ prioritize PLM.

Branch Agreement Loss (Resolves Challenge 3)

$$\mathcal{L}_{\text{agree}} = \frac{1}{N} \sum_i \left\| \text{sm}(\mathbf{z}_{\text{EGNN},i}) - \text{sm}(\mathbf{z}_{\text{GT},i}) \right\|_2^2$$

($\text{sm}(\cdot)$ = softmax)

Composite Training Objective

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{focal}}(\gamma=2.0, \alpha^+ \approx 5.4) \\ & + \lambda_{\text{bio}} \mathcal{L}_{\text{bio}} + \lambda_{\text{agree}} \mathcal{L}_{\text{agree}} \end{aligned}$$

Threshold Selection Protocol

Threshold θ^* locked on validation fold (argmax val-F1) before blind test evaluation — no data leakage [6].

Codebase and Engineering

- Extended from open-source GTE-PPIS PyTorch/DGL codebase [6].
- **ESM-2** (650M, esm2_t33_650M_UR50D) [8]: 1280D residue embeddings extracted and pre-cached offline.
- FFM: Gated mode (149D combined) vs. concatenation baseline (277D).
- Training: Residual connections (all 10 EGNN + 4 GT blocks), LayerNorm in projection, gradient clipping (norm 1.0).

Datasets and Evaluation Protocol

- **Train:** Train_335 (334 valid protein chains). 5-fold CV with fixed seed (SEED=2024).
- **Bound Test:** Test_60 (60 chains), Test_315-28 (287 chains).
- **Unbound Test:** UBtest_31-6 (31 chains).
- Metrics: AUROC, AUPRC, MCC, F1, Recall, Precision.

Model Variants Compared

- **No-ESM2 Baseline:** Structural features only ($\gamma = 2.0$).
- **Unsupervised Gate:** ESM-2 gated without RSA supervision.
- **RSA-Supervised Gate:** Full biophysical supervision ($\lambda_{\text{gate}} = 0.1$).

Benchmark	Model Variant	AUROC	AUPRC	MCC	F1	Recall	Precision
Test_60 (Bound)	GTE-PPIS Published ^[6]	0.8730	0.6110	0.5000	0.5820	0.6110	0.5570
	DHEG Published ^[7]	0.9000	0.6860	0.5800	0.6480	0.6670	0.6300
	No-ESM2 Baseline (Ours)	0.8537	0.5518	0.4500	0.5319	0.5734	0.5050
	Unsupervised Gate (Ours)	0.8236	0.5056	0.4125	0.5089	0.5395	0.4855
	RSA-Supervised Gate (Ours)	0.8116	0.4863	0.3874	0.4925	0.5708	0.4372
Test_315-28 (Bound)	DHEG Published ^[7]	—	0.6510	0.5510	—	—	—
	GTE-PPIS Published ^[6]	—	0.5980	0.5110	—	—	—
	No-ESM2 Baseline (Ours)	0.8585	0.5182	0.4354	0.5189	0.6819	0.4190
	Unsupervised Gate (Ours)	0.8117	0.4381	0.3601	0.4597	0.6104	0.3690
	RSA-Supervised Gate (Ours)	0.8029	0.4351	0.3497	0.4492	0.6447	0.3458

Bold = best value in that benchmark group. Wang et al. ^[6] (Table 3) and Li et al. ^[7] (Table 2) published only AUPRC and MCC for Test_315-28 (threshold-independent metrics); remaining cells shown as —.

Benchmark	Model Variant	AUROC	AUPRC	MCC	F1	Recall	Precision
UBtest_31-6 (Unbound)	DHEG Published [7]	0.8210	0.4050	0.3700	0.4520	0.5680	0.3760
	GTE-PPIS Published [6]	—	0.3430	0.3200	—	—	—
	No-ESM2 Baseline (Ours)	0.7611	0.3115	0.2713	0.3542	0.3404	0.3714
	Unsupervised Gate (Ours)	0.8191	0.4182	0.3706	0.4454	0.4644	0.4385
	RSA-Supervised Gate (Ours)	0.8159	0.4248	0.3694	0.4495	0.5024	0.4106

Key Finding 1 – Unbound Generalisation

Gated ESM-2 vs. No-ESM2 on UBtest_31-6:

AUPRC: 0.3115 → 0.4248 (+36.4%)

MCC: 0.2713 → 0.3694 (+36.2%)

PLM embeddings provide conformational invariance that pure coordinate-based GNNs lose during backbone displacement.

Key Finding 2 – Avoiding PLM Collapse

DHEG [7]: naive ESM-2 concat collapsed

MCC from **0.580** → **0.315** on Test_60.

Our gated FFM maintains **MCC $\approx$ 0.41**, validating that learned gating is essential for safe PLM integration.

DHEG evaluated UBtest on 25 chains (single split); all Ours variants: UBtest_31-6, 31 chains, 5-fold CV averages, validation-locked thresholds.

Finding 1 – Unbound Generalization Gain

- On UBtest_31-6, the coordinate-only model (No-ESM2) drops to **AUPRC 0.3115, MCC 0.2713** due to conformational shifting.
- Gated ESM-2 fusion achieves **AUPRC 0.4248 (+36.4%)** and **MCC 0.3694 (+36.2%)**.
- PLM sequence representations provide invariance when 3D backbone coordinates shift during binding displacement.

Finding 2 – Bound Structure Redundancy

- On static bound crystals (Test_60), No-ESM2 achieves higher AUPRC (0.5518 vs. 0.5056).
- Bound coordinates directly encode interface geometry; 1280D PLM features add slight redundancy on static data.

Finding 3 – Preventing Catastrophic PLM Collapse

- In DHEG [7], unconstrained ESM-2 concatenation collapsed MCC from **0.580 down to 0.315**.
- Our learned FFM gating protects geometric signals, maintaining **MCC $\approx$ 0.41** on Test_60.
- Confirms that adaptive gating is essential for multi-modal protein learning.

Confounder Audit in RSA Run

- The preliminary RSA-supervised run altered three settings simultaneously ($\lambda_{\text{gate}} = 0.1$, $\gamma = 0.0$, residual connections).
- A controlled ablation with $\gamma = 2.0$ fixed will isolate the pure RSA supervision gain.

Summary of Contributions

- **Biophysically-supervised FFM:** Scalar gate g_i balances 1280D ESM-2 against classical features, supervised continuously by RSA and B-factor.
- **Unbound generalisation:** Gated ESM-2 achieves **+0.1133 AUPRC**, **+0.0981 MCC** on UBtest_31-6 over coordinate-only baselines.
- **Structural discriminability preserved:** Gate-controlled integration prevents PLM feature collapse (avoiding DHEG [7] degradation from 0.580 $\rightarrow$ 0.315).
- **Branch agreement regularisation:** Prevents latent feature drift between parallel EGNN and Graph Transformer encoders.

Future Work and Next Steps

1. Controlled RSA ablation: fix $\gamma = 2.0$ and isolate the pure λ_{gate} supervision effect.
2. Multi-seed evaluation ($N \geq 3$) to report statistical confidence intervals.
3. Partner-specific conditioning to extend prediction to dynamic docking complexes.

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